psymetadata: Intelligence example

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```r
library(metaselection)
library(metadat)
library(psymetadata)
library(metafor)
```

```
## Loading required package: Matrix

## Loading required package: numDeriv

##
## Loading the 'metafor' package (version 4.8-0). For an
## introduction to the package please type: help(metafor)
```

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x tidyr::expand() masks Matrix::expand()
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## x tidyr::pack()   masks Matrix::pack()
## x tidyr::unpack() masks Matrix::unpack()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(here)
```

```
## here() starts at C:/GitHub/metaselection
```

```r
devtools::load_all()
```

```
## i Loading metaselection
```

Intelligence Meta-analysis

```
dat_ex <- nuijten2020

dat_ex <- dat_ex |>
  rename(esid = es_id, studyid = study_id) |>
  mutate(
    d = yi,
    Va = vi,
    Var_d = Va,
    sd_d = sqrt(Var_d),
    var_d = Var_d,
    p_onesided = 1 - pnorm(d / sd_d)
  )

length(unique(dat_ex$studyid)); length(unique(dat_ex$esid))
```

```
## [1] 1917
```

```
## [1] 2443
```

Mean Only

Unadjusted estimate: mean only

```
ex_unadj_mean <- rma.uni(
  data = dat_ex,
  yi = d,
  sei = sd_d,
  method = "ML"
)
```

```
## Warning: 4 studies with NAs omitted from model fitting.
```

```
ex_unadj_mean
```

```
##
## Random-Effects Model (k = 2439; tau^2 estimator: ML)
##
## tau^2 (estimated amount of total heterogeneity): 0.0513 (SE = 0.0020)
## tau (square root of estimated tau^2 value):      0.2265
## I^2 (total heterogeneity / total variability):   99.75%
## H^2 (total variability / sampling variability):  399.21
##
## Test for Heterogeneity:
## Q(df = 2438) = 32270.7666, p-val < .0001
##
## Model Results:
```

```
##
## estimate      se     zval    pval   ci.lb   ci.ub
##   0.2639  0.0054  49.2515  <.0001  0.2534  0.2744  ***
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

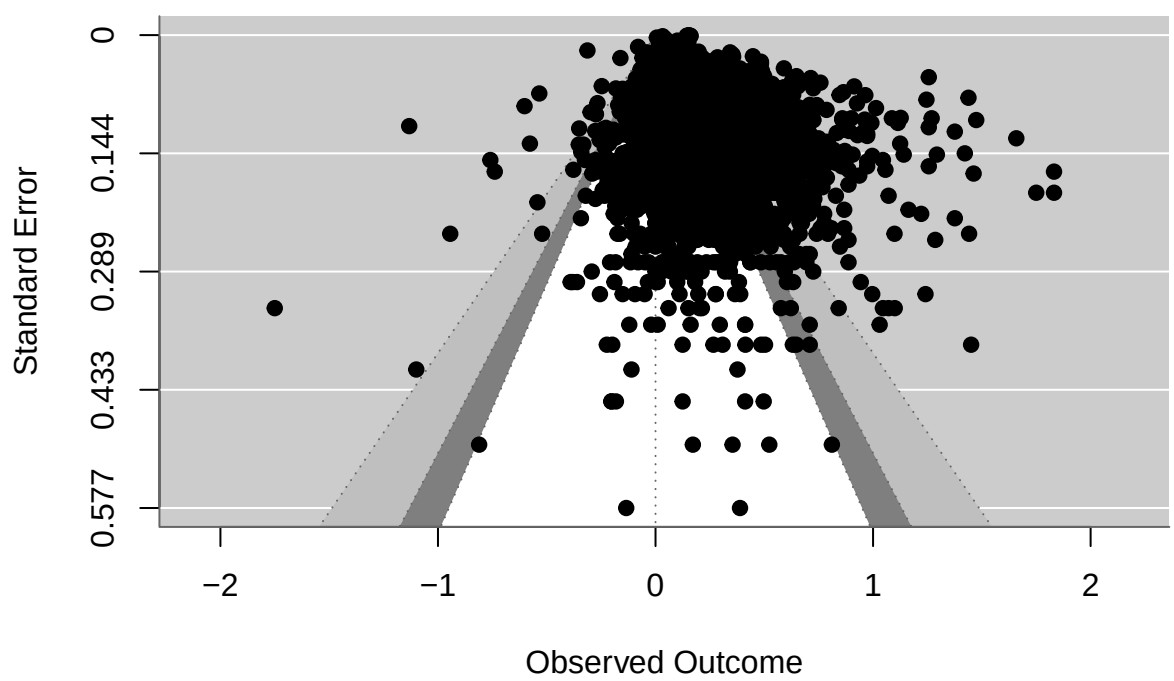
```
ex_unadj_mean_cil <- ex_unadj_mean$tau2 - 1.96*ex_unadj_mean$se.tau2
ex_unadj_mean_ciu <- ex_unadj_mean$tau2 + 1.96*ex_unadj_mean$se.tau2
ex_unadj_mean_cil; ex_unadj_mean_ciu
```

```
## [1] 0.04748448
```

```
## [1] 0.05513954
```

Forest plot

```
funnel(ex_unadj_mean, refline = 0, level = c(90, 95, 99))
```



CHE estimate: mean only

```r
V_mat <- vcalc(
    vi = vi,
    cluster = studyid,
    obs = esid,
    data = dat_ex,
    rho = 0.5,
    sparse = TRUE
  )

ex_che <- rma.mv(yi = yi,
                 V = V_mat,
                 random = ~ 1 | studyid / esid,
                 data = dat_ex,
                 sparse = TRUE,
                 digits = 3) |>
    robust(cluster = studyid, clubSandwich = TRUE)
```

```
## Warning: 4 rows with NAs omitted from model fitting.
```

```r
ex_che
```

```
##
## Multivariate Meta-Analysis Model (k = 2439; method: REML)
##
## Variance Components:
##
##            estim   sqrt  nlvls  fixed        factor
## sigma^2.1  0.023  0.150   1913     no       studyid
## sigma^2.2  0.029  0.169   2439     no  studyid/esid
##
## Test for Heterogeneity:
## Q(df = 2438) = 31931.064, p-val < .001
##
## Number of estimates:   2439
## Number of clusters:    1913
## Estimates per cluster: 1-14 (mean: 1.27, median: 1)
##
## Model Results:
##
## estimate     se¹    tval¹       df¹   pval¹  ci.lb¹  ci.ub¹
##    0.258  0.006   44.606   1776.77   <.001   0.247   0.269   ***
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## 1) results based on cluster-robust inference (var-cov estimator: CR2,
##    approx t-test and confidence interval, df: Satterthwaite approx)
```

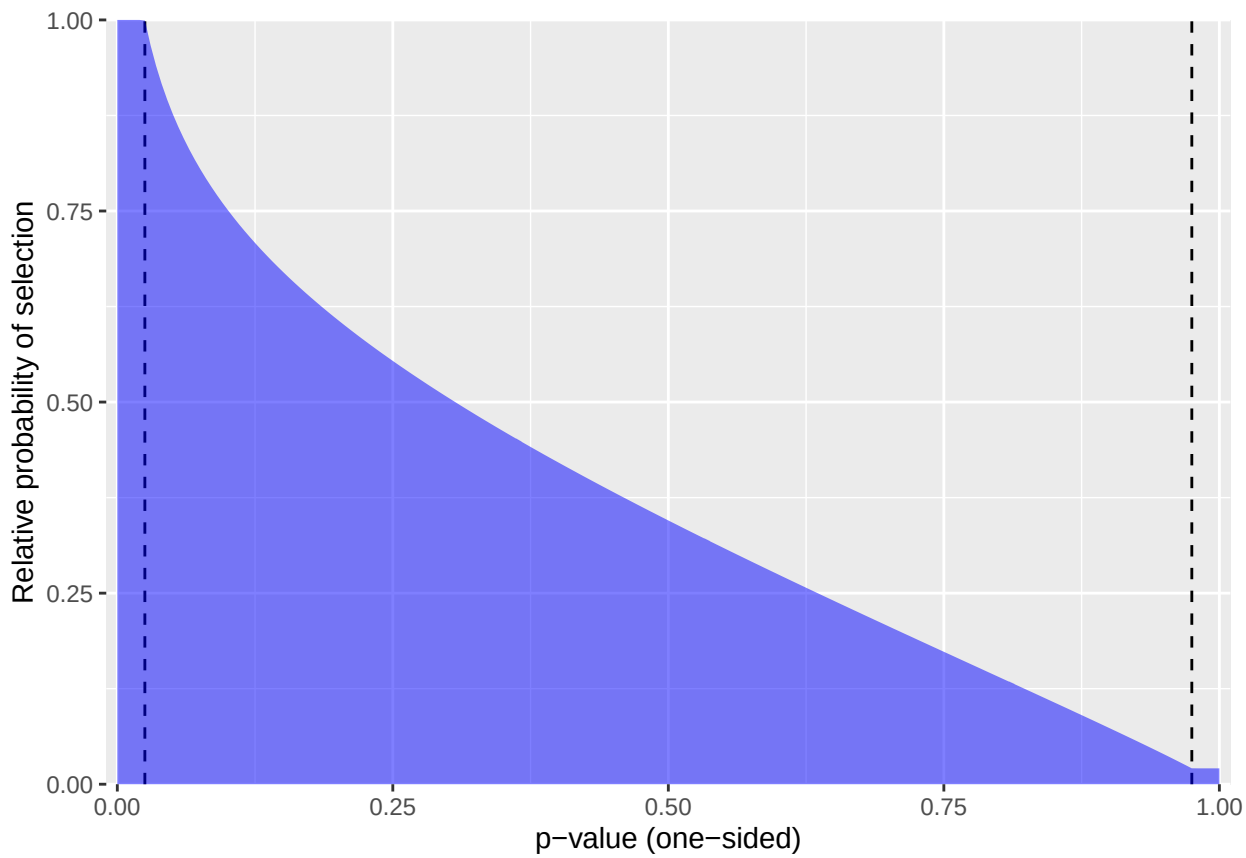
Adjusted estimate in metaselection: Beta model mean only

```r
ex_adj_ms_beta_mean <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  pi = p_onesided,
  cluster = studyid,
  selection_type = "beta",
  steps = c(.025,.975),
  make_sandwich = TRUE
)
ex_adj_ms_beta_mean
```

```
##    param     Est     SE  p_value  CI_lo   CI_hi
##     beta -0.0945 0.0526 7.23e-02 -0.197 0.00854
##     tau2  0.1323 0.0158       NA  0.105 0.16715
##  lambda1  0.8464 0.0324 1.33e-05  0.785 0.91238
##  lambda2  1.9041 0.0858 2.79e-46  1.743 2.07999
```

Selection plot: Beta model mean only

```r
selection_plot(ex_adj_ms_beta_mean)
```



Adjusted estimate in metaselection: Step 3PSM model mean only

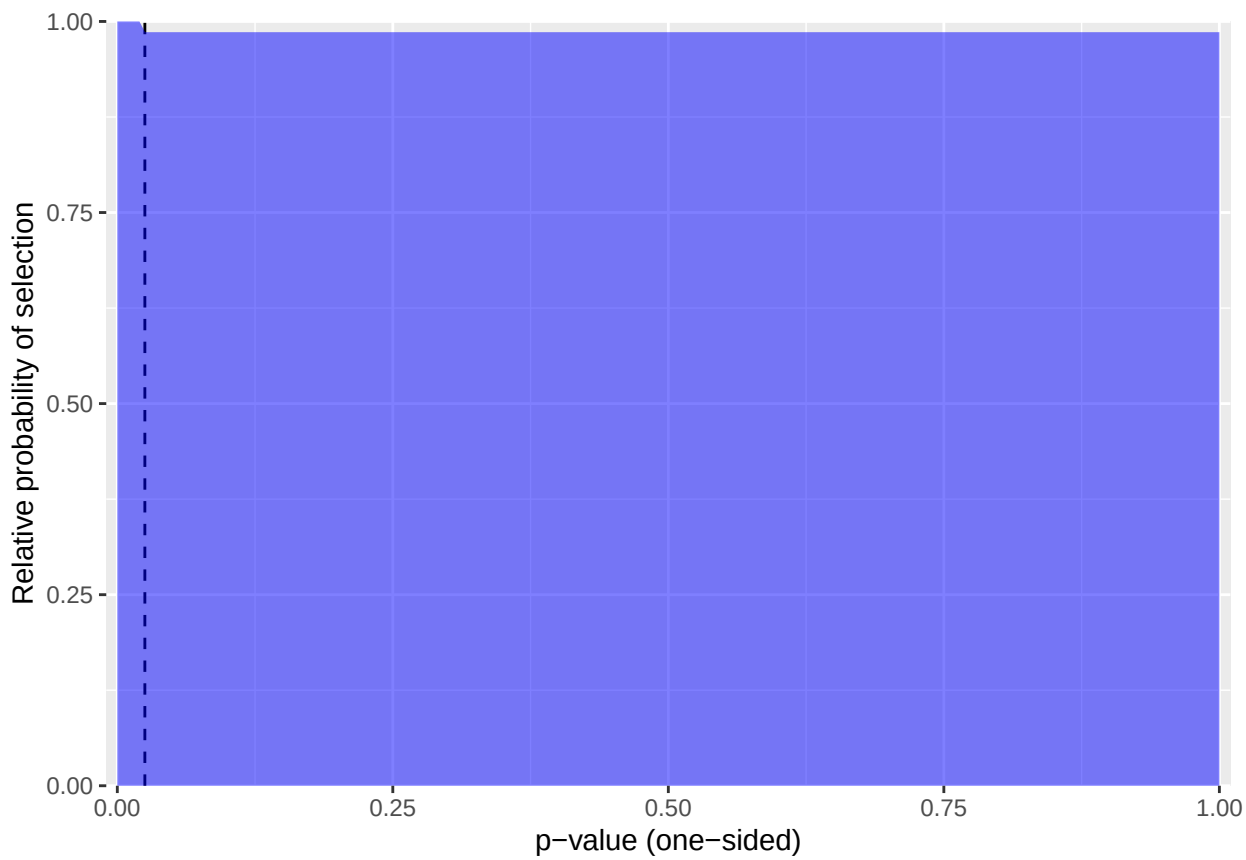
```
ex_adj_ms_step3_mean <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  cluster = studyid,
  selection_type = "step",
  steps = 0.025
)

ex_adj_ms_step3_mean
```

```
##    param    Est      SE   p_value  CI_lo  CI_hi
##     beta 0.2626 0.00994 7.78e-154 0.2431 0.2821
##     tau2 0.0513 0.00358        NA 0.0448 0.0589
##  lambda1 0.9860 0.07445  8.52e-01 0.8503 1.1432
```

Selection plot: Step 3 PSM model mean only

```
selection_plot(ex_adj_ms_step3_mean)
```



Adjusted estimate in metaselection: Step 4PSM model mean only

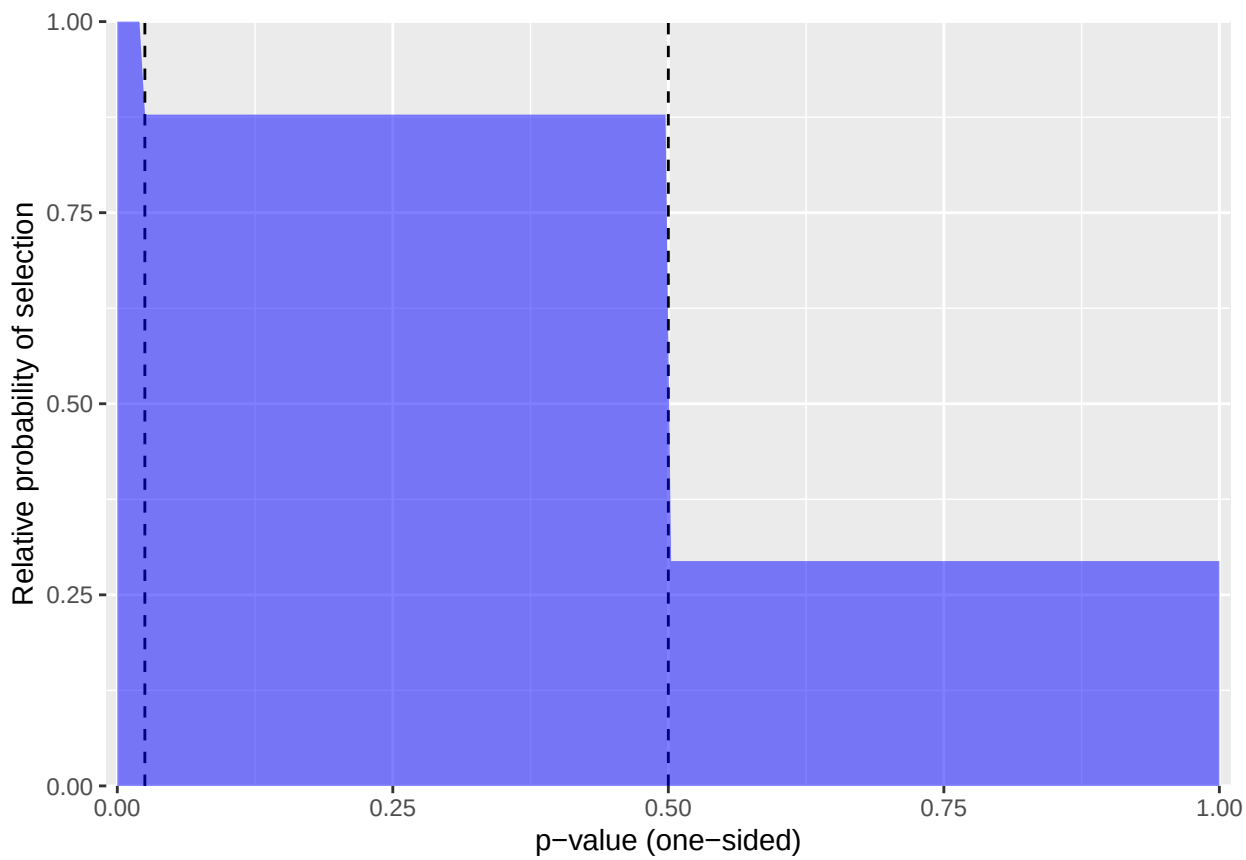
```
ex_adj_ms_step4_mean <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  cluster = studyid,
  selection_type = "step",
  steps = c(0.025, .5)
)

ex_adj_ms_step4_mean
```

```
##    param    Est      SE  p_value  CI_lo  CI_hi
##     beta 0.1698 0.01597 2.02e-26 0.1385 0.2011
##     tau2 0.0716 0.00639       NA 0.0601 0.0853
##  lambda1 0.8783 0.06245 6.80e-02 0.7640 1.0097
##  lambda2 0.2942 0.04552 2.64e-15 0.2172 0.3984
```

Selection plot: Step 4PSM model mean only

```
selection_plot(ex_adj_ms_step4_mean)
```



Moderator

Unadjusted estimate: moderator

```
ex_unadj_mod <- rma.uni(
  data = dat_ex,
  yi = d,
  sei = sd_d,
  method = "ML",
  mods = ~ jrnl_impact
)
```

```
## Warning: 550 studies with NAs omitted from model fitting.
```

```
ex_unadj_mod
```

```
##
## Mixed-Effects Model (k = 1893; tau^2 estimator: ML)
##
## tau^2 (estimated amount of residual heterogeneity):     0.0471 (SE = 0.0021)
## tau (square root of estimated tau^2 value):             0.2170
## I^2 (residual heterogeneity / unaccounted variability): 96.24%
## H^2 (unaccounted variability / sampling variability):   26.61
## R^2 (amount of heterogeneity accounted for):            0.03%
##
## Test for Residual Heterogeneity:
## QE(df = 1891) = 17466.2232, p-val < .0001
##
## Test of Moderators (coefficient 2):
## QM(df = 1) = 0.3036, p-val = 0.5817
##
## Model Results:
##
##              estimate      se     zval    pval    ci.lb   ci.ub
## intrcpt        0.2550  0.0060  42.6889  <.0001   0.2433  0.2667  ***
## jrnl_impact    0.0001  0.0002   0.5510  0.5817  -0.0003  0.0005
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ex_unadj_mod_cil <- ex_unadj_mod$tau2 - 1.96*ex_unadj_mod$se.tau2
ex_unadj_mod_ciu <- ex_unadj_mod$tau2 + 1.96*ex_unadj_mod$se.tau2
ex_unadj_mod_cil; ex_unadj_mod_ciu
```

```
## [1] 0.04302507
```

```
## [1] 0.05114433
```

Adjusted estimate in metaselection: Beta model moderator

```r
ex_adj_ms_beta_mod <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  pi = p_onesided,
  cluster = studyid,
  selection_type = "beta",
  steps = c(.025,.975),
  make_sandwich = TRUE,
  mean_mods = ~ jrnl_impact
)
```

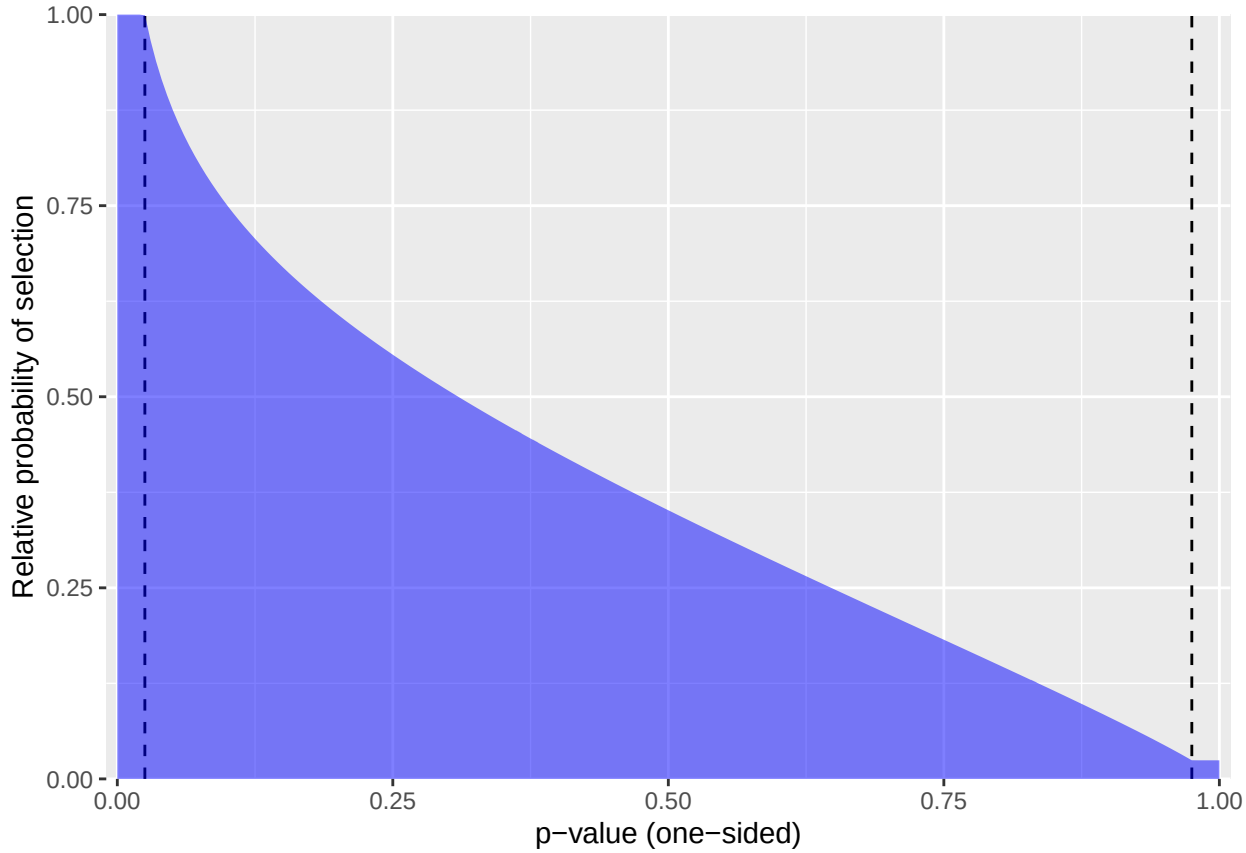
```
## Warning in optimx.check(par, optcfg$ufn, optcfg$ugr, optcfg$uhess, lower, : Parameters or bounds appe
##   This can cause poor performance in optimization.
##   It is important for derivative free methods like BOBYQA, UOBYQA, NEWUOA.
```

```r
ex_adj_ms_beta_mod
```

```
##             param       Est       SE  p_value     CI_lo    CI_hi
##  beta_(Intercept) -0.065186 0.057304 2.55e-01 -1.77e-01 0.047128
##  beta_jrnl_impact  0.000257 0.000135 5.69e-02 -7.60e-06 0.000522
##              tau2  0.116360 0.015806       NA  8.92e-02 0.151855
##           lambda1  0.841846 0.037006 8.99e-05  7.72e-01 0.917592
##           lambda2  1.856196 0.099136 5.14e-31  1.67e+00 2.061034
```

Selection plot: Beta model moderator

```r
selection_plot(ex_adj_ms_beta_mod)
```



Adjusted estimate in metaselection: Step 3PSM model moderator

```
ex_adj_ms_step3_mod <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  cluster = studyid,
  selection_type = "step",
  steps = 0.025,
  mean_mods = ~ jrnl_impact
)
```

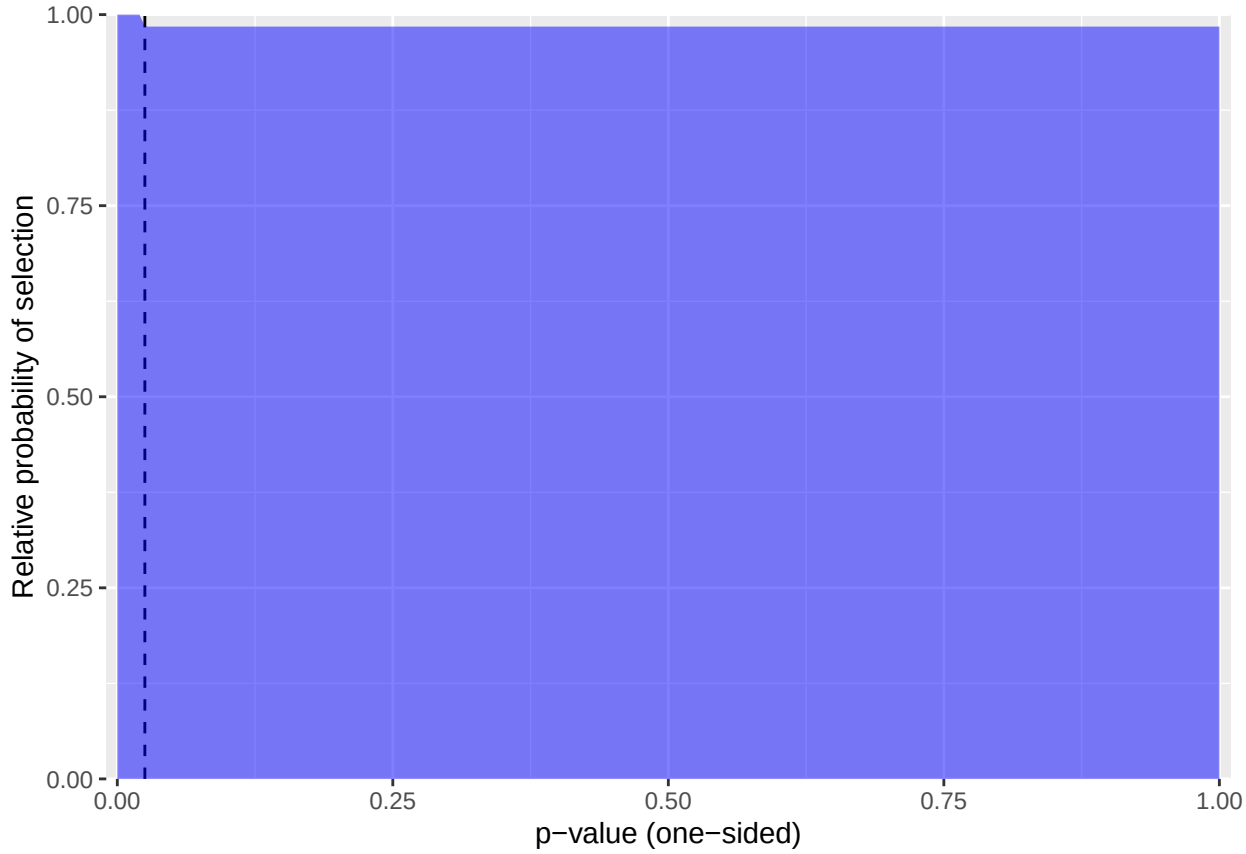
```
## Warning in optimx.check(par, optcfg$ufn, optcfg$ugr, optcfg$uhess, lower, : Parameters or bounds appe
##   This can cause poor performance in optimization.
##   It is important for derivative free methods like BOBYQA, UOBYQA, NEWUOA.
```

```
ex_adj_ms_step3_mod
```

```
##             param      Est       SE   p_value     CI_lo    CI_hi
##  beta_(Intercept) 0.253499 1.06e-02 1.78e-125  2.33e-01 0.274352
##  beta_jrnl_impact 0.000121 7.68e-05  1.15e-01 -2.94e-05 0.000272
##              tau2 0.047089 3.50e-03        NA  4.07e-02 0.054466
##           lambda1 0.984339 8.34e-02  8.52e-01  8.34e-01 1.162160
```

Selection plot: Step 3 PSM model moderator

```
selection_plot(ex_adj_ms_step3_mod)
```



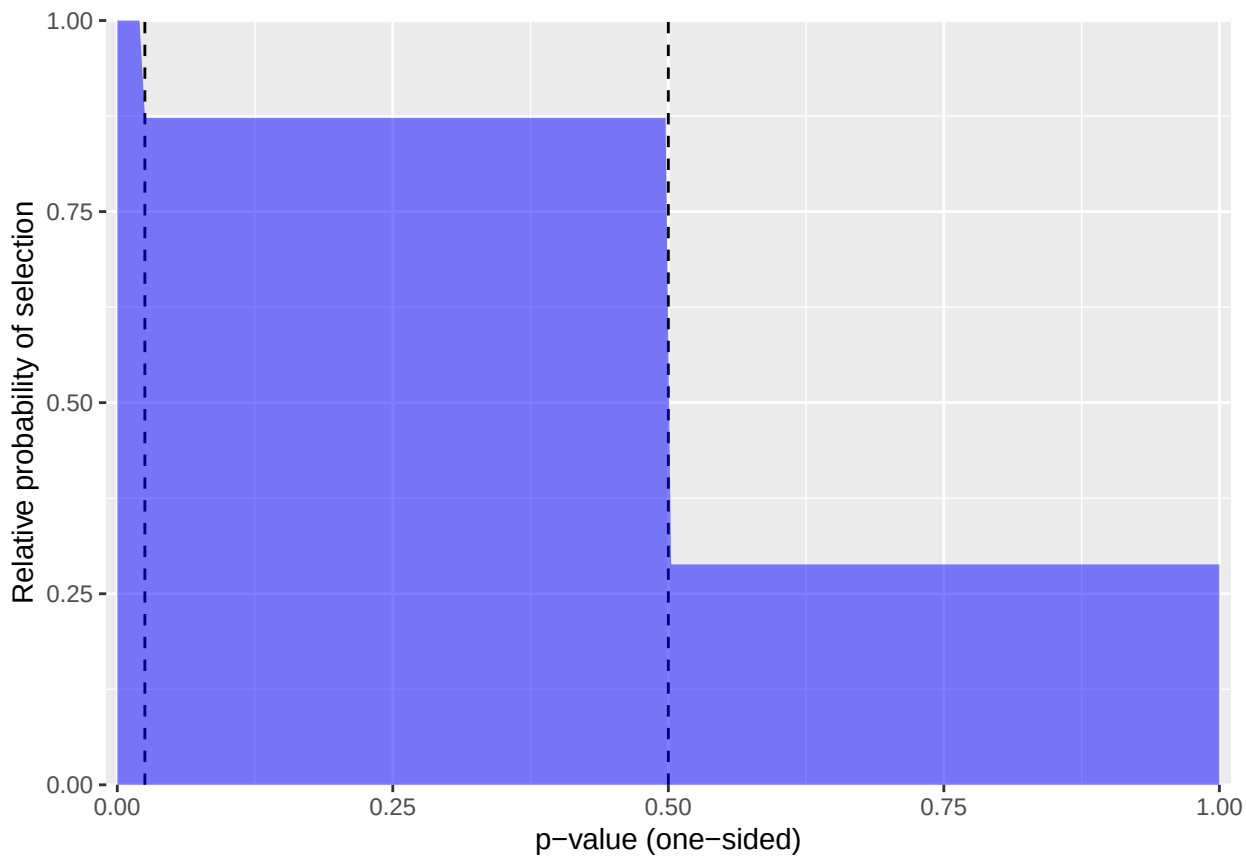
Adjusted estimate in metaselection: Step 4PSM model moderator

```
ex_adj_ms_step4_mod <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  cluster = studyid,
  selection_type = "step",
  steps = c(0.025, .5),
  mean_mods = ~ jrnl_impact
)
```

```
## Warning in optimx.check(par, optcfg$ufn, optcfg$ugr, optcfg$uhess, lower, : Parameters or bounds appe
##   This can cause poor performance in optimization.
##   It is important for derivative free methods like BOBYQA, UOBYQA, NEWUOA.
```

```r
ex_adj_ms_step4_mod
```

```
##             param      Est       SE  p_value     CI_lo    CI_hi
##  beta_(Intercept) 0.162454 1.81e-02 2.95e-19  1.27e-01 0.197948
##  beta_jrnl_impact 0.000169 8.95e-05 5.94e-02 -6.69e-06 0.000344
##              tau2 0.066171 6.36e-03       NA  5.48e-02 0.079897
##           lambda1 0.872454 7.02e-02 9.01e-02  7.45e-01 1.021589
##           lambda2 0.288405 5.13e-02 2.78e-12  2.03e-01 0.408742
```

Selection plot: Step 4PSM model moderator

```r
selection_plot(ex_adj_ms_step4_mod)
```